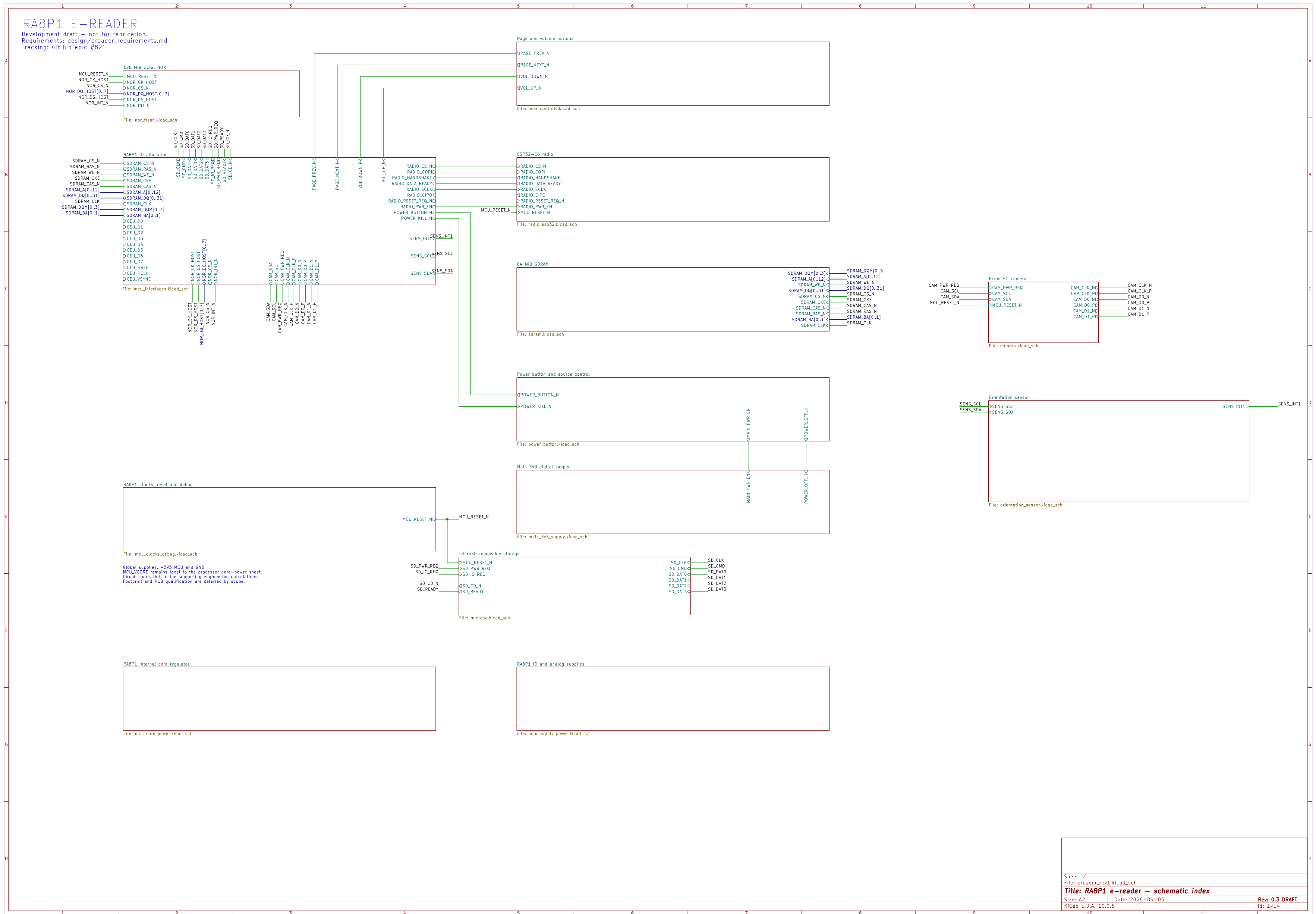
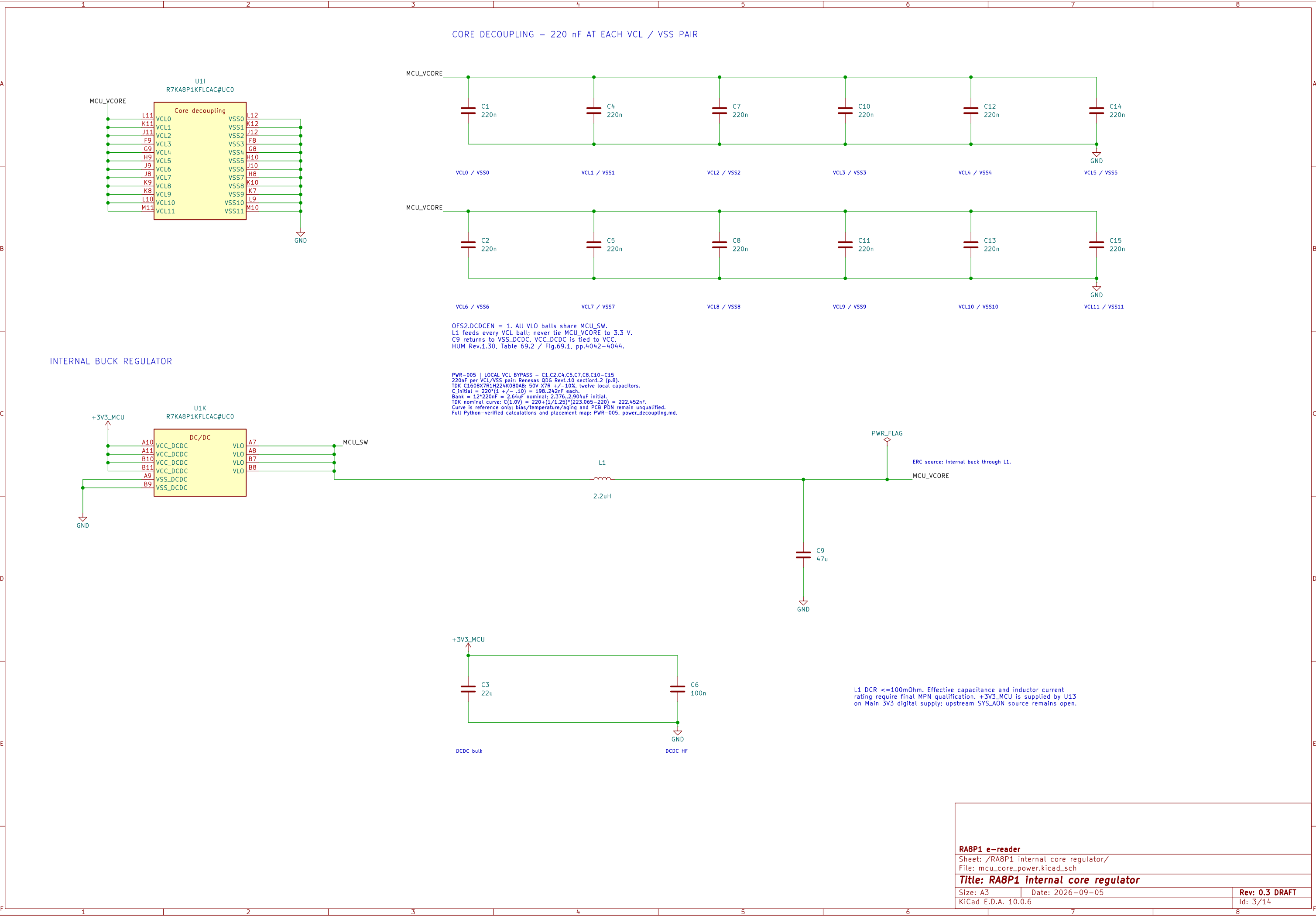
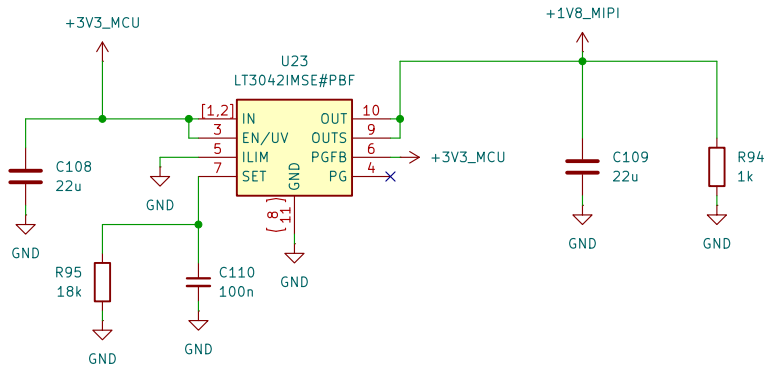


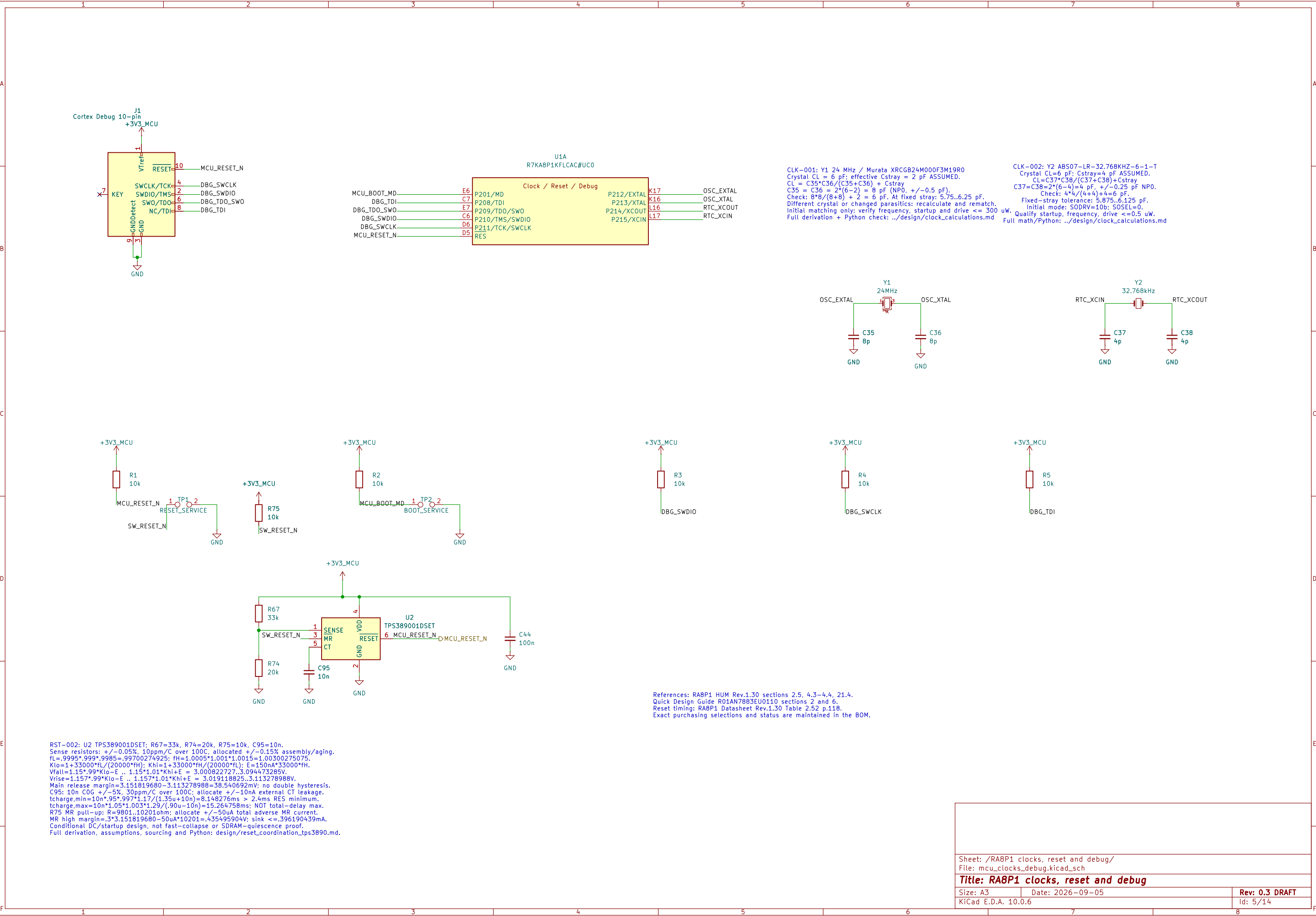
Development draft – not for fabrication.
Requirements: design/ereader_requirements.md
Tracking: GitHub epic #821.





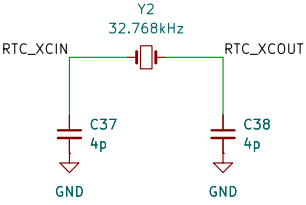
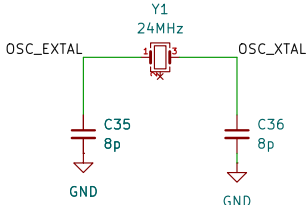


Id: 4/14



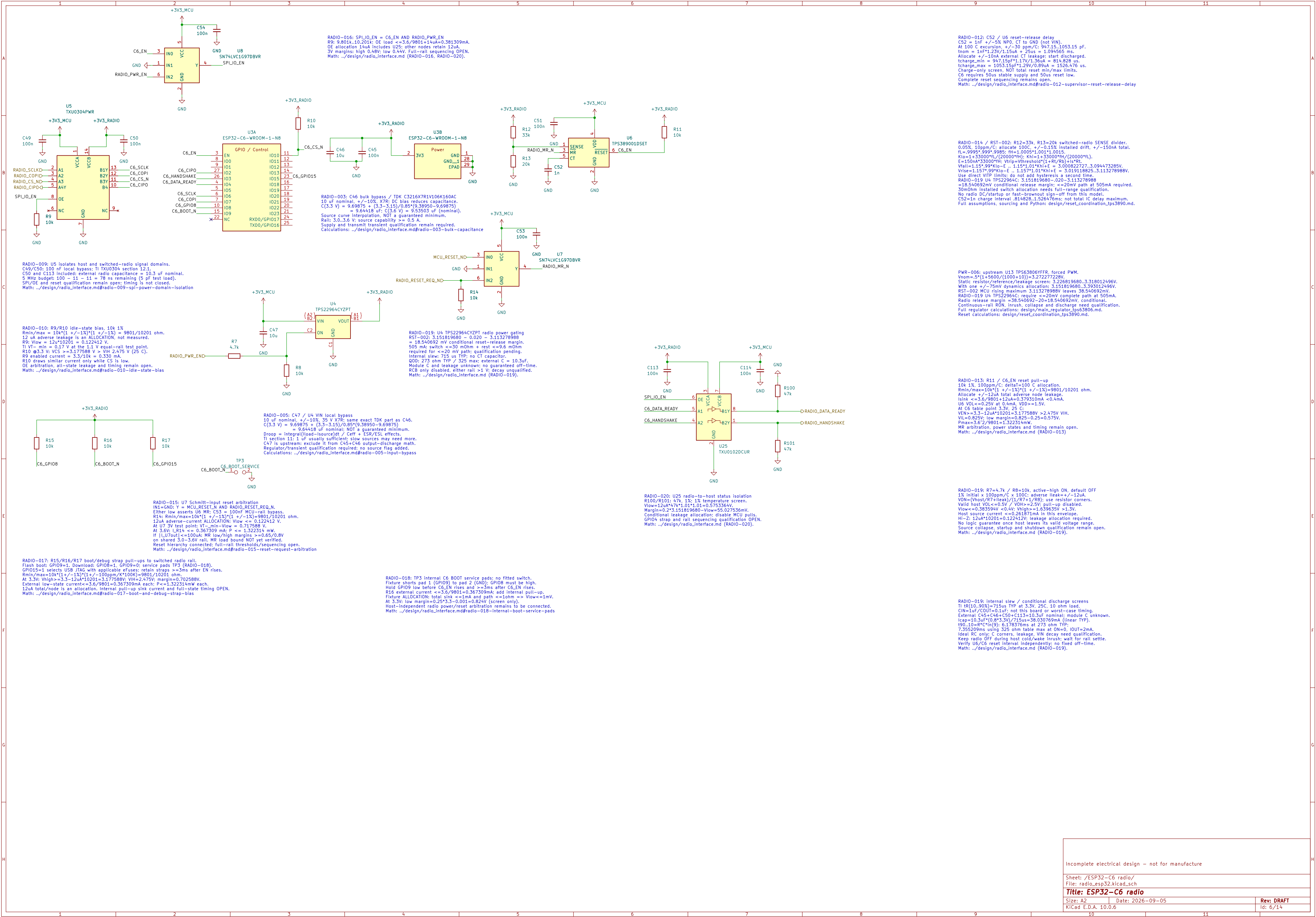
CLK-001: Y1 24 MHz / Murata XRCGB24M000F3M19R0
Crystal CL = 6 pF: effective Cstray = 2 pF ASSUMED.
CL = C35*C36/(C35+C36) + Cstray
C35 = C36 = 2*(6-2) = 8 pF (NP0, +/-0.5 pF).
Check: 8*8/(8+8) + 2 = 6 pF. At fixed stray: 5.75..6.25 pF.
Different crystal or changed parasitics: recalculate and rematch.
Initial matching only: verify frequency, startup and drive <= 300 uW.
Full derivation + Python check: ../design/clock_calculations.md

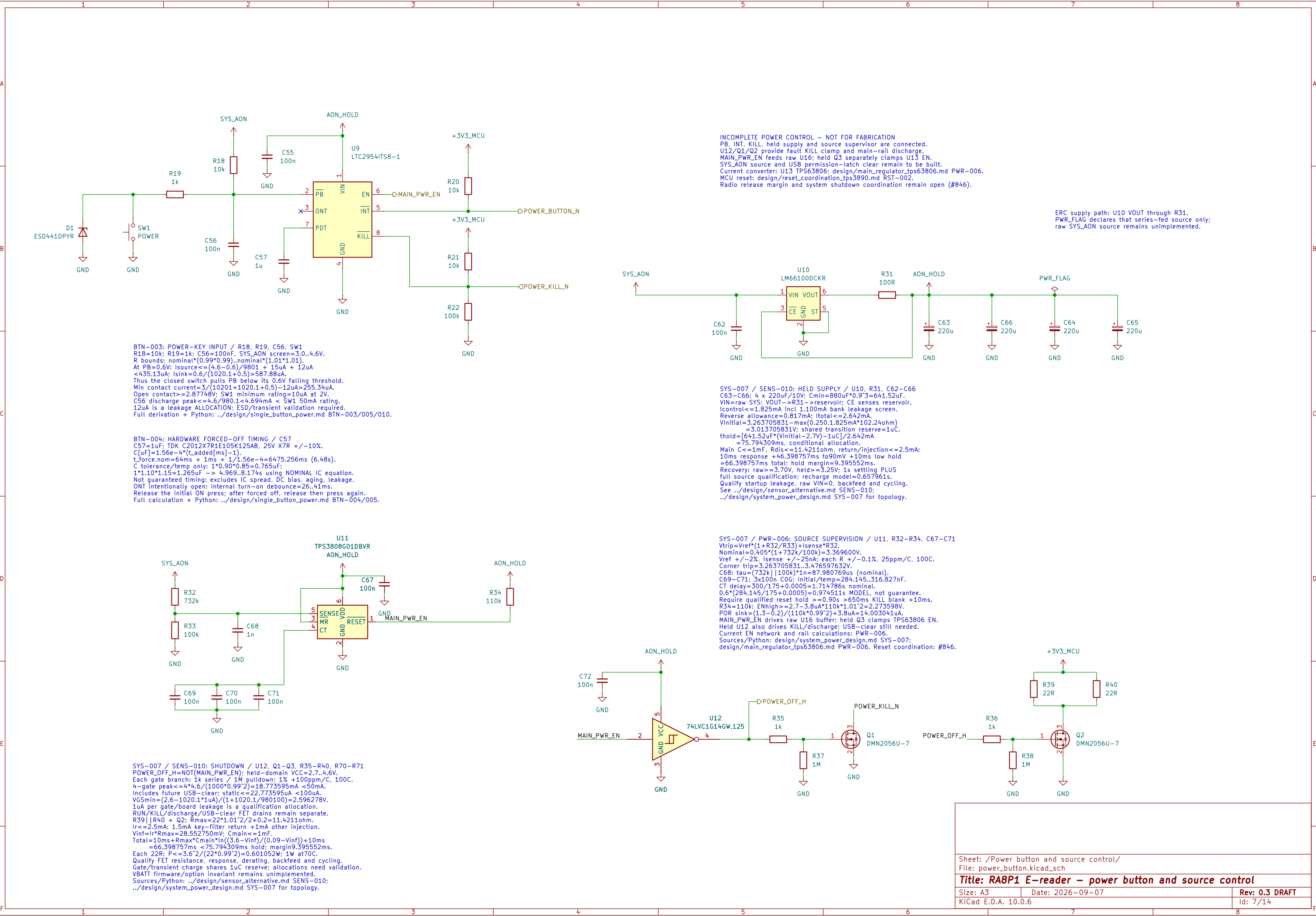
CLK-002: Y2 ABS07-LR-32.768KHZ-6-1-T
Crystal CL=6 pF: Cstray=4 pF ASSUMED.
CL=C37*C38/(C37+C38)+Cstray
C37=C38=2*(6-4)=4 pF, +/-0.25 pF NP0.
Check: 4*4/(4+4)+4=6 pF.
Fixed-stray tolerance: 5.875..6.125 pF.
Initial mode: SODRV=10b; SOSEL=0.
Qualify startup, frequency, drive <=0.5 uW.
Full math/Python: ../design/clock_calculations.md

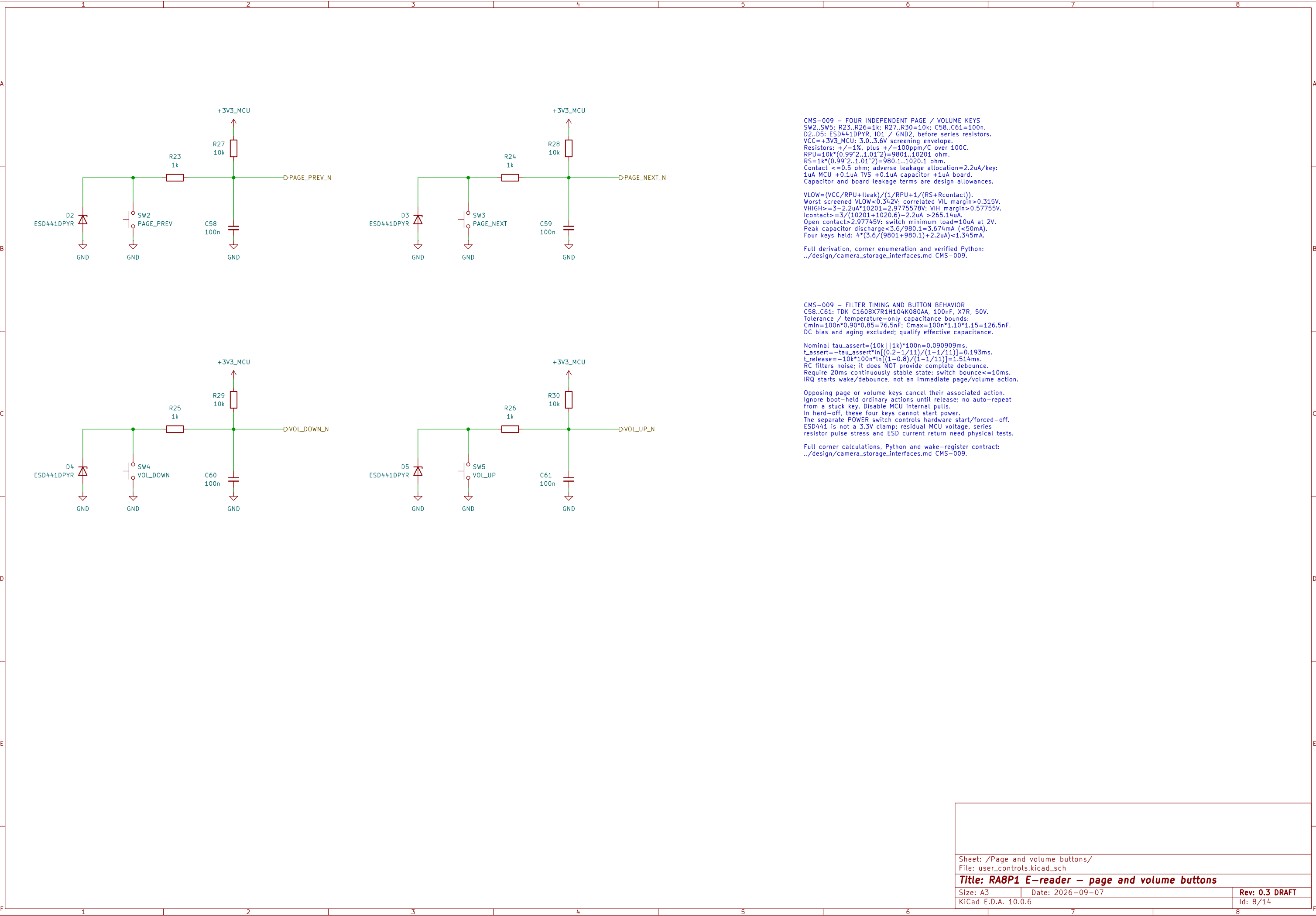


References: RA8P1 HUM Rev.1.30 sections 2.5, 4.3-4.4, 21.4.
Quick Design Guide R01AN7883EU0110 sections 2 and 6.
Reset timing: RA8P1 Datasheet Rev.1.30 Table 2.52 p.118.
Exact purchasing selections and status are maintained in the BOM.

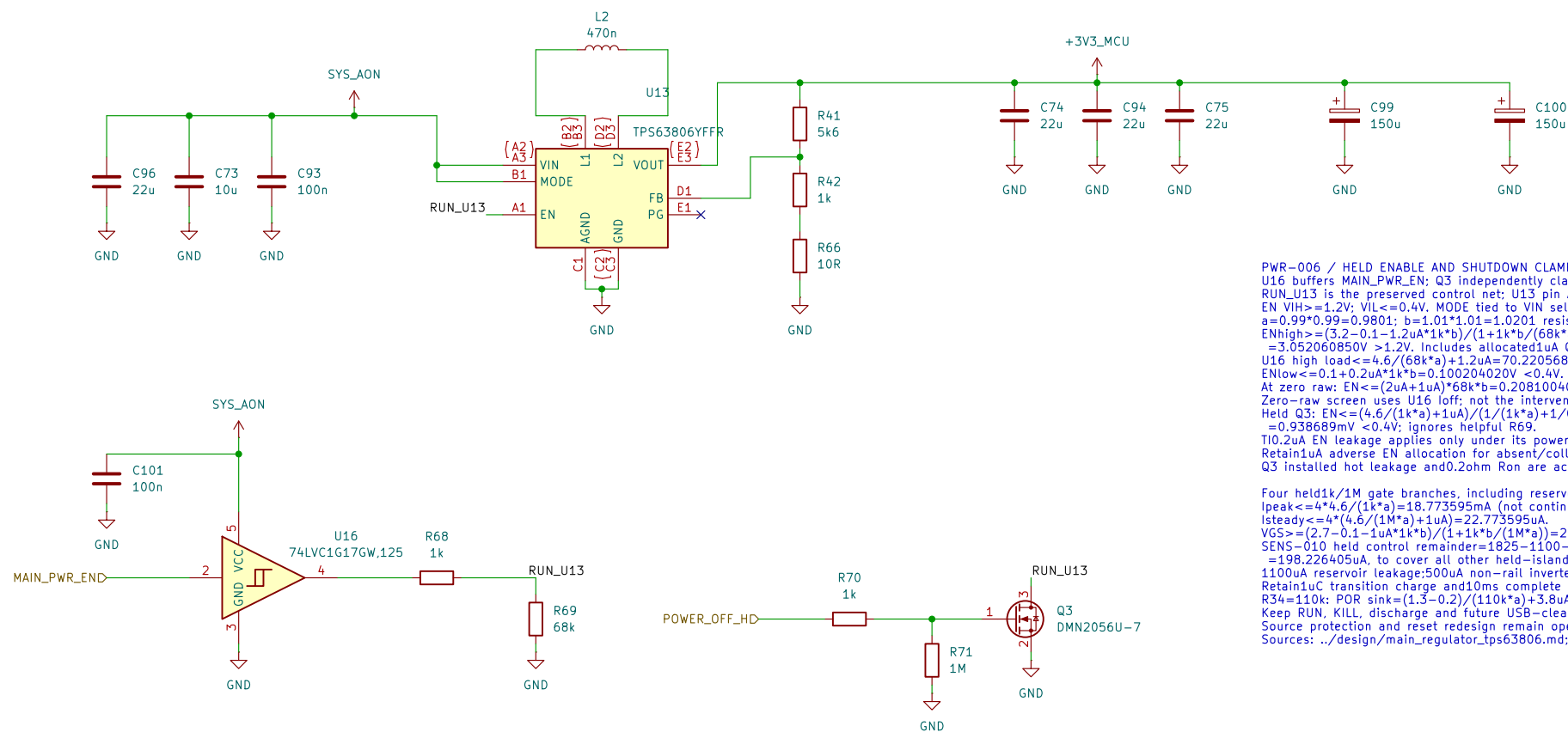
RST-002: U2 TPS389001DSET; R67=33k, R74=20k, R75=10k, C95=10n.
Sense resistors: +/-0.05%, 10ppm/C over 100C, allocated +/-0.15% assembly/aging.
fL=.9995*.999*.9985=.99700274925; fH=1.0005*1.001*1.0015=1.00300275075.
KLo=1+33000*fL/(20000*fH); KHi=1+33000*fH/(20000*fL); E=150nA*33000*fH.
Vfall=1.15*.99*KLo-E .. 1.15*1.01*KHi+E = 3.000822727..3.094473285V.
Vrise=1.157*.99*KLo-E .. 1.157*1.01*KHi+E = 3.019118625..3.113278988V.
Main release margin=3.151819680-3.113278988=38.540692mV; no double hysteresis.
C95: 10n C0G +/-5%, 30ppm/C over 100C; allocate +/-10nA external CT leakage.
tcharge.min=10n*.95*.997*1.17/(1.35u+10n)=8.148276ms > 2.4ms RES minimum.
tcharge.max=10n*1.05*1.003*1.29/(.90u-10n)=15.264758ms; NOT total-delay max.
R75 MR pull-up: R=9801..10201ohm; allocate +/-50uA total adverse MR current.
MR high margin=.3*3.151819680-50uA*10201=.435495904V; sink <=.396190439mA.
Conditional DC/startup design, not fast-collapse or SDRAM-quiescence proof.
Full derivation, assumptions, sourcing and Python: design/reset_coordination_tps3890.md.







MAIN DIGITAL SUPPLY: TPS63806 / PWR-006
3.272277 V nominal; forced PWM; held shutdown clamp.
Divider and bypass calculations: design/main_regulator_tps63806.md
Reset coordination (#846), source protection and qualification remain open.
DEVELOPMENT SCHEMATIC – NOT FOR POWER-UP OR FABRICATION.



```

PWR-006 / HLD ENABLE AND SHUTDOWN CLAMP
U16 buffers MAIN_PWD_EN; Q3 independently clamps EN during raw collapse.
RUN_U13 is the preserved control net; U13 pin A1 is now EN.
EN VIH>=1.2V; VIL<=0.4V. MODE tied to VIN selects forced PWM.
=0.99+0.99*0.9801; b=1.01*1.01+1.0201 (resistor factors)
= (3.2-0.1-1.2uA)*1k*b/(1-0.99)
= 3.052060850uA. Included allclocks_U13 off leakage.
U16 high load = 4.6/(68k*1) + 1.2uA/70.220568uA <100uA.
ENlow = 0.1+0.2uA*1k*b = 0.100204020V <0.4V.
At zero raw: EN<=(2uA+1uA)*68k*b = 0.208100400V.
Zero-raw screen uses U16 loff; not the intervening undefined region.
Held Q3: EN<= (4.6/(1k*1)+1uA)/(1/(1k*1)+1/0.2ohm)
= 0.9386mV <0.4V; ignores held U16.
T10.2uA EN leakage applies only under its powered test conditions.
RetainU1A adverse EN allocation for absent/collapsing raw.
Q3 installed hot leakage and0.2ohm Rn are acceptance allocations.

```

```

Four held1k/1M gate branches, including reserved USB-clear:
Ipeak<=4*(4+6/(1k^a))=18.773595mA (not continuous ratings).
Isteady<=4*(4+6/(1M^a)+1A)=22.773595mA.
VG5=(2.7-0.1/(1n+1k^b))*(1+1k^b/(1M^a))=2.566278V.
SENS=high-level control, remainder=862.5mA=22.773595
=198.226450uA, to cover all other held-island loads.
1100uA reservoir leakage;500uA non-rail inverter-input allocation.
RetainLCU transition charge and10ms complete response allocation.
R34=110K; POR sink=(1.3-0.2)/(110K^a)+3.8uA=14.003041uA.
KILL=KILL pin, no charge and no current source separate.
Source protection and reset redesign remain open; no power-up approval.
Sources.../design/main_regulator_tps63806.m; SENS=010 revision.

```

PWR-Vom / FEEDBACK AND LOCAL PASSIVES
 $V_{nom} = 5.00 \times (1 + 5600 / (1000 + 10)) = 3.27227228 \text{ V}$
R41: 0.02%, 5ppm/C; R42: 0.01%, 2ppm/C; R66: 1%, 200ppm/C.
Installed boards: initial/TRC(1000) "aging" assembly, then ohms.
 $R41 = 5600 \times (1 + (-0.0002)) \times (1 + (-0.0005)) \times (1 + (-0.001))$
 $\times (1 + (-0.0005)) \div (-0.02) = 5587.6629237, 5612.349243 \text{ ohm}$
 $R42 = 1000 \times (1 + (-0.0001)) \times (1 + (-0.0002)) \times (1 + (-0.001))$
 $\times (1 + (-0.0005)) \div (-0.01) = 998.190970, 1001.810970 \text{ ohm}$
 $R66 = 10 \times (1 + (-0.01)) \times (1 + (-0.02)) \times (1 + (-0.01)) \div (-0.05)$
 $= 9.954980, 10.455020 \text{ ohm}$ one combined aging/assembly 1%.
 $V_{in} = 5.00 \times (1 + 141680 / (R41 \times R42 \times R66)) = 5.00 \times 1.01 \times 1.01 \times 1.01$
 $\times V_{in} = 5.05 \times (1 + R41 \times R42 \times R66) = 5.05 \times (1 + 100nA \times R41 \times \text{max}$
 $\text{Vstatic} = 3.226819680, 3.318012496 \text{ V (FPWM only)}$
One combined + / -75mV regulation/ripple/transient/routing allowance:
 $V_{complete} = 3.151819680, 3.393012496 \text{ V}$: qualification required.
SDRAM static VOH margin = $2.4 - 0.73 = 3.93012496 \text{ V} = 24.891253 \text{ mV}$.
This is not measured S/I/noise headroom or reset qualification.

12: 470nH +/-20%; catalog spread alone is insufficient.
Screen incoming0.412..0.518uH; allocated operating+/-10%
= +/-3708..0.5698uH versus T137..0.57uH. Quality hot/current.
C74/C75/C94: *322uF*0.8*0.85^0.6=6.2928uF > T121uF
C96: 22uF*0.8*0.85^0.6=8.976uF > T14uF input minimum.
The capacitor combined resistance is not guaranteed.
Samsung typical output bias retention at3.393012496V-90.221554%.
Input bias, aging, mounted impedance and startup need verification.
Full assumptions, sources and executable Python: click this note.

PWR-006 / LDA, STORAGE AND SHUTDOWN
Allocation: 1.8A operating; 2.25A cold start screen, not measured load.
Pwr=3.393012496e+1,8=6.107424932E+3W
Thermal allocated Tjmax efficiency: Pin=84.33229991W; lin@3.2V=2,544759372A
Losses = 2.68078670W; loss = 0.678602499W
Tj90C efficiency: loss=0.678602499W.
ATJ50C=70C+78.8C/0.678602499W=123.474C.
2.5 board theta -JA is not final PCB thermal resistance.
Pin=3.393012496e+1,8=6.107424932E+3W
Converter limit, source limit, thermal and thermal envelope need validation.

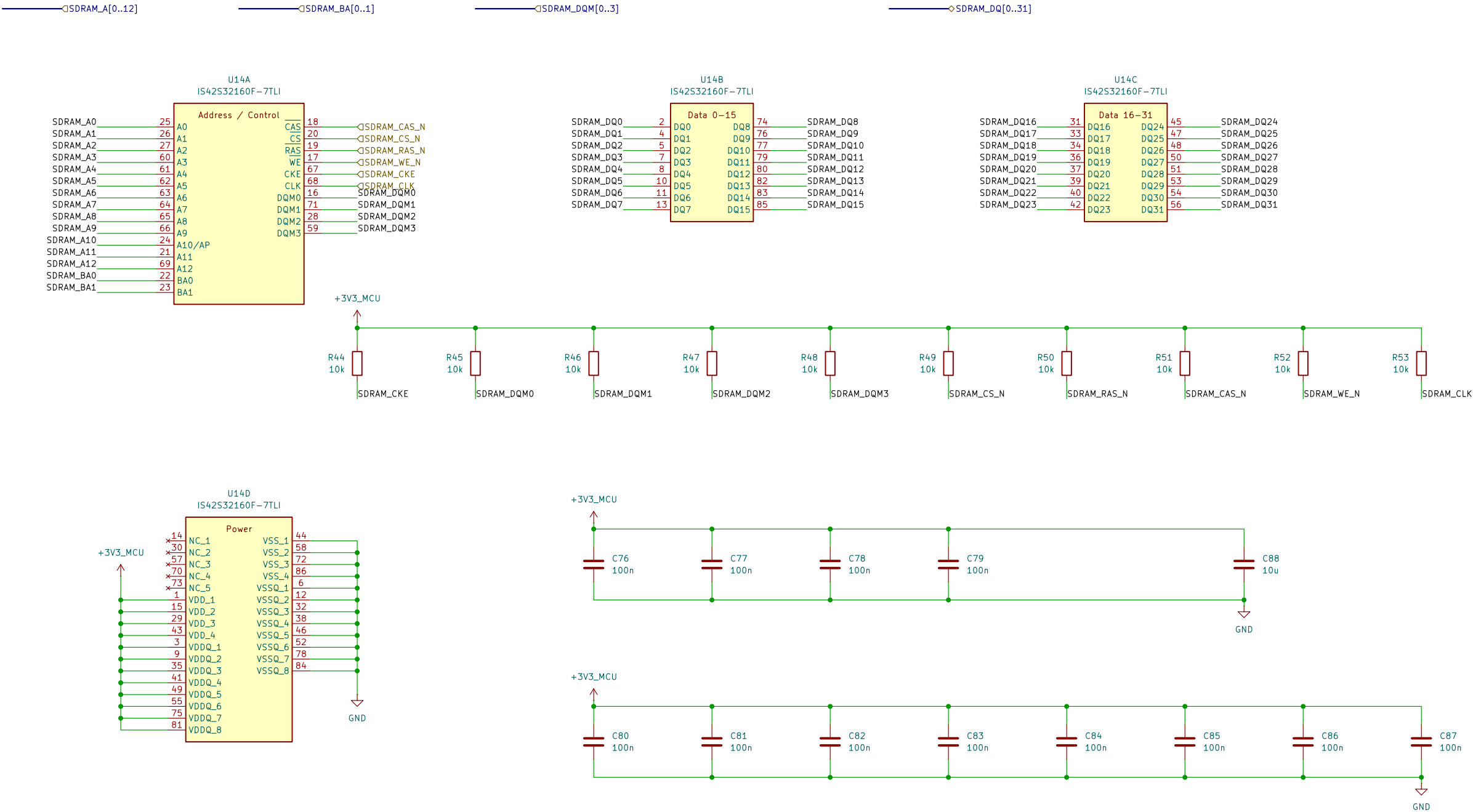
PWR=006 storage checkpoint (nominal, not effective minimum):
Direct main469.21uF + C43 through FB1 10uF + key filters0.4uF
=479.61uF cold; radio-on add±0.2uF =>489.81uF.
C99/C100 remain±150uF each. C94 adds22uF to the output.
Old bootstrap/private VCC/VyF capacitors were not main-rail storage.
Whole-main ceiling remains1mF; reconcile later peripheral additions.
Cout lower acceptance and distributed PDN/startup are separate checks.

```

SENS-010 discharge revision: Rmax1+0.42110hm: return/injection<=2.5mA.
Includes1.5mA key return +1mA other backfeed: Vinf=0.028552750V.
t=1mF+11.42110hm*ln((3.6-Vinf)/(0.09-Vinf))+10ms+10ms
=66.3987575ms total, below475.794309ms screen.
Includes10ms delay+10ms ramp/discharge+10ms delay+10ms below90ms.
Held budget:1.825mA control +0.817mA reverse;Ic transition reserve.
198.226405uA control remainder must cover other held-island loads.
SYS_AON sensor protection, loaded startup and fast faults remain open.
RSN_002 / RADIO-019: release margin remains conditional on rail/path bounds.
No motor power or sensor control or communication connection are complete.
Full assumptions: ../design/main_regulator_tps63806.mt
shutdown revision/Python: ../design/sensor_alternative.md SENS-010.

```

64 MiB SDRAM | ISSI IS42S32160F-7TLI
Signal and reset—default circuits | CMS-010.
Startup, SI/timing and hardware qualification remain open.



CMS-010 | SDRAM power and bypass
VDD and VDDQ share +3V3_MCU (3.0 to 3.6 V).
C76–C79: 4 x 100 nF for the four VDD pins.
C80–C87: 8 x 100 nF for the eight VDDQ pins.
C88: 10 uF local bulk; Cnom = 12 x 0.1 + 10 = 11.2 uF.
Included in the PWR-003 1 mF whole-rail limit.
Effective C, placement and PDN impedance require qualification.

CMS-010 | R44–R53: 10k reset defaults
Rail 3.0..3.6V: +/-1%, 100ppm/C. |dT| <=100C.
Rmin=10k*0.99*0.99=9801 ohm.
Rmax=10k*1.01*1.01=10201 ohm.
Ioff=5uA SDRAM+1uA MCU+1uA board=7uA.
VHIGHmin=3.0-7uA*10201=2.928593V (>2.0V VIH).
Board leakage and temperature screen require qualification.

CMS-010 | Pull load and power states
Sink leakage: 5uA SDRAM+1uA board=6uA.
Isink=3.6/9801+6uA=0.373309mA (<0.374mA).
Pmax=3.6^2/9801=1.322314mW per pull.
All ten LOW: 10*3.6/9801=3.673095mA rail:
10*3.6^2/9801=13.223140mW resistor heat.
Reserve 4mA within the 250mA SDRAM allocation.
CKE/DQM and CS_N default HIGH; CLK HIGH is not a clock.
Switched-rail pulls; self-refresh actively holds CKE LOW.
Startup, dynamic I/O, SI and shutdown qualification OPEN.

Sheet: /64 MiB SDRAM/
File: sdram.kicad_sch

Title: RA8P1 E-reader - 64 MiB SDRAM

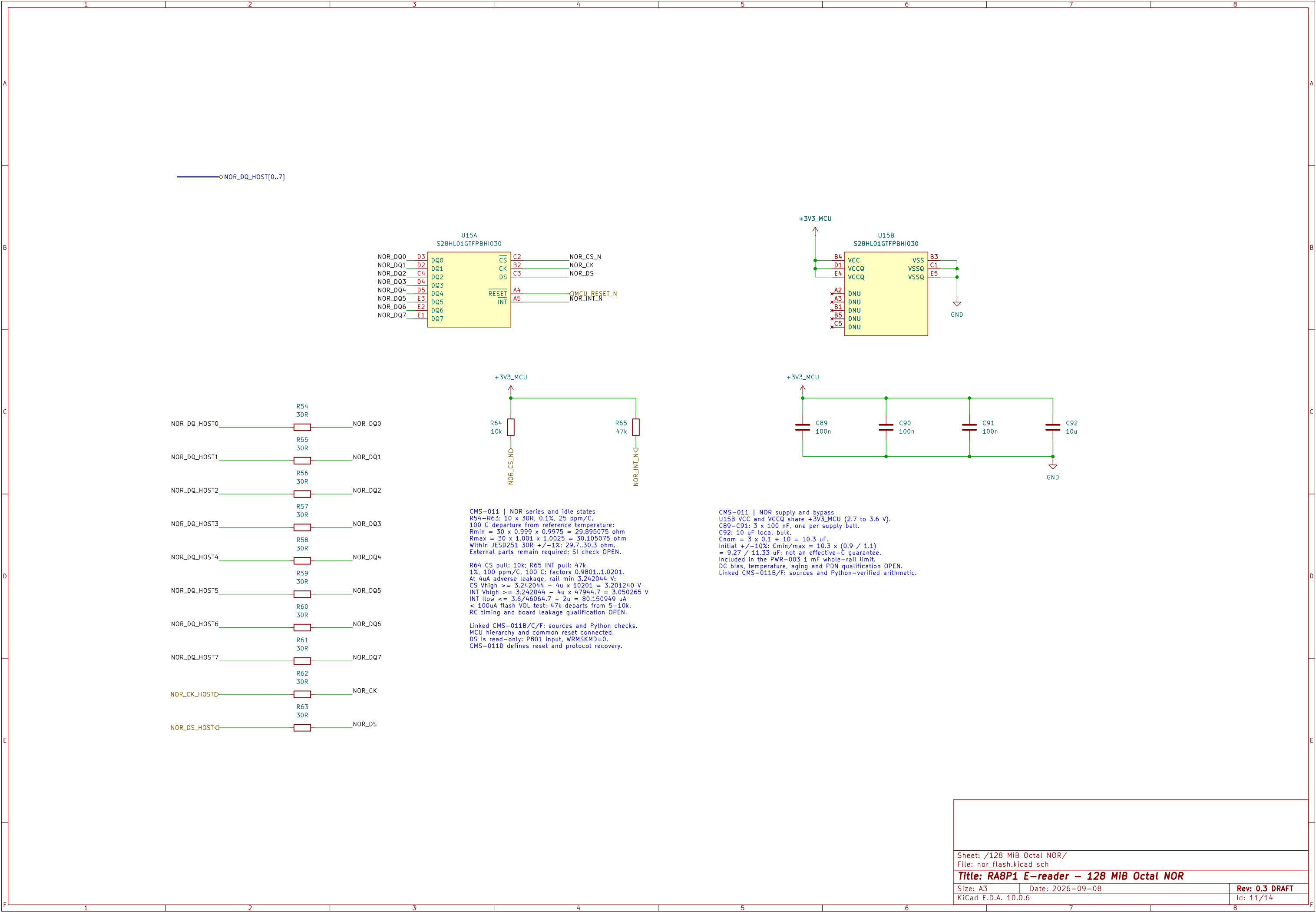
Size: A3

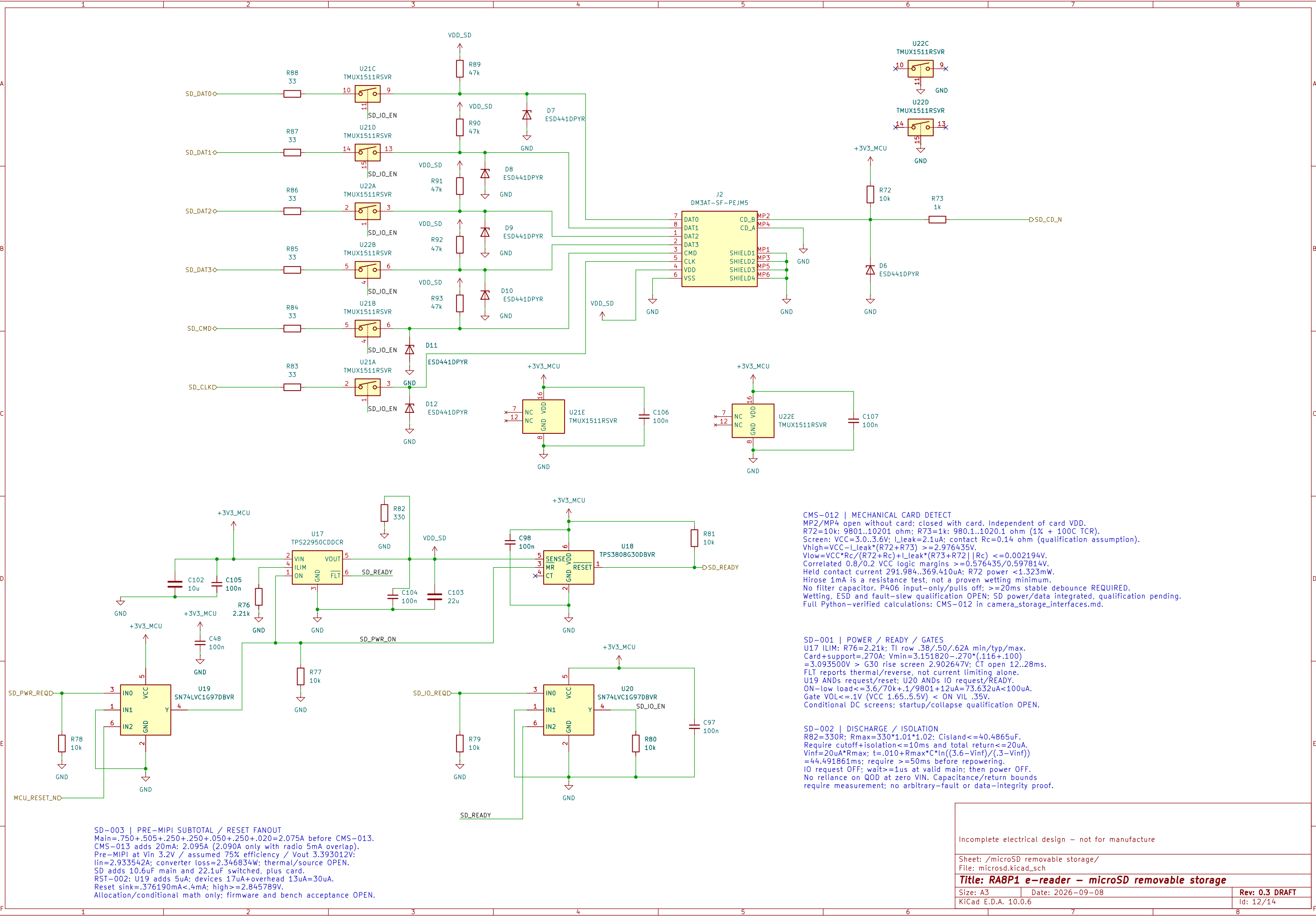
Date: 2026-09-08

Rev: 0.3 DRAFT

KiCad E.D.A. 10.0.6

Id: 10/14





SD-003 | PRE-MIPI SUBTOTAL / RESET FANOUT
Main=.750+.505+.250+.250+.050+.250+.020=2.075A before CMS-013.
CMS-013 adds 20mA: 2.095A (2.090A only with radio 5mA overlap).
Pre-MIPI at Vin 3.2V / assumed 75% efficiency / Vout 3.393012V:
Iin=2.933542A; converter loss=2.346834W; thermal/source OPEN.
SD adds 10.6uF main and 22.1uF switched, plus card.
RST-002: U19 adds 5uA; devices 17uA+overhead 13uA=30uA.
Reset sink=.376190mA<.4mA; high>=2.845789V.
Allocation/conditional math only; firmware and bench acceptance OPEN.

CMS-012 | MECHANICAL CARD DETECT
MP2/MP4 open without card; closed with card. Independent of card VDD.
R72=10k; 9801..10201 ohm; R73=1k; 9801..1020.1 ohm (1% + 100C TCR).
Screen: VCC=3.0..3.6V; ILeak=2.1uA; contact Rc=0.14 ohm (qualification assumption).
Vhigh=VCC-ILeak*(R72+R73) >=2.976435V.
Vlow=VCC*Rc/(R72+Rc)+ILeak*(R73+R72) | Rc) <=0.002194V.
Correlated 0.8/0.2 VCC logic margins >=0.576435/0.597814V.
Held contact current 291.984..369.410uA; R72 power <1.323mW.
Hiros 1mA is a resistance test, not a proven wetting minimum.
No filter capacitor. P406 input-only/pulls off; >=20ms stable debounce REQUIRED.
Wetting, ESD and fault-slew qualification OPEN; SD power/data integrated, qualification pending.
Full Python-verified calculations: CMS-012 in camera_storage_interfaces.md.

SD-001 | POWER / READY / GATES
U17 ILIM: R76=2.21k; TI row .38/.50/.62A min/typ/max.
Card+support=.270A; Vmin=3.151820-.270*(.116+.100)
=3.093500V > G30 rise screen 2.902647V; CT open 12..28ms.
FLT reports thermal/reverse, not current limiting alone.
U19 ANDs request/reset; U20 ANDs IO request/READY.
ON-low load<=3.6/70k+.1/9801+12uA=73.632uA<100uA.
Gate VOL<=.1V (VCC 1.65..5.5V) < ON VIL .35V.
Conditional DC screens; startup/collapse qualification OPEN.

SD-002 | DISCHARGE / ISOLATION
R82=330R; Rmax=330*1.01*1.02; Cisland<=40.4865uF.
Require cutoff+isolation<=10ms and total return<=20uA.
Vinf=20uA*Rmax; t=.010+Rmax*C*ln((3.6-Vinf)/(.3-Vinf))
=44.491861ms; require >=50ms before repowering.
IO request OFF; wait>=1us at valid main; then power OFF.
No reliance on QOD at zero VIN. Capacitance/return bounds
require measurement; no arbitrary-fault or data-integrity proof.

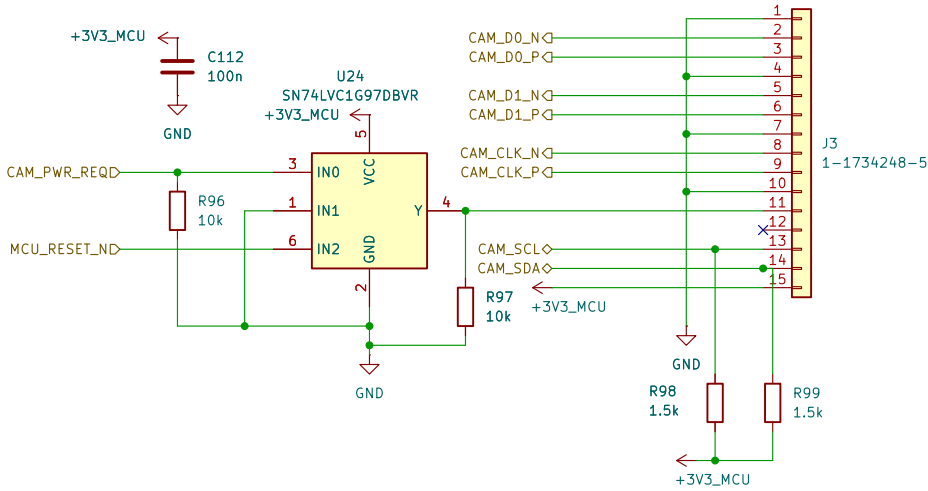
Incomplete electrical design – not for manufacture

Sheet: /microSD removable storage/
File: microsd.kicad_sch

Title: RA8P1 e-reader – microSD removable storage

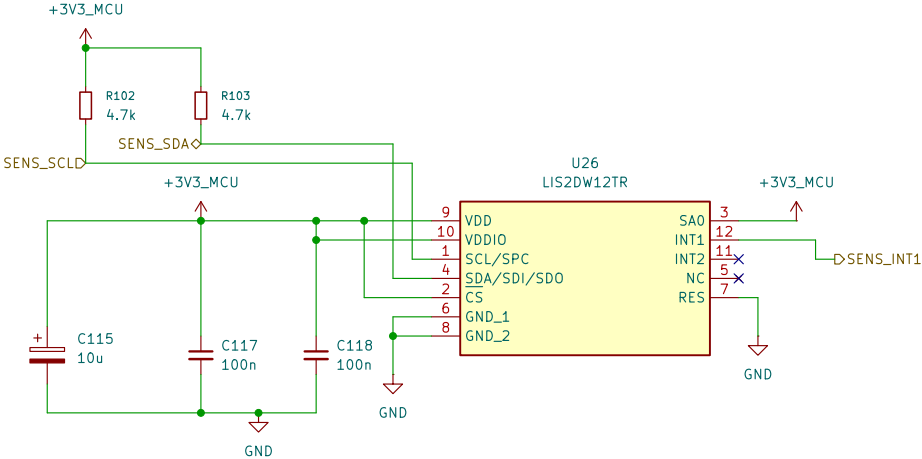
Size: A3 Date: 2026-09-08
KiCad E.D.A. 10.0.6

Rev: 0.3 DRAFT
Id: 12/14



CMS-014 / CMS-015: Pcam 5C, 2-lane CSI-2 to VIN0
U24: PWUP = CAM_PWR_REQ AND MCU_RESET_N.
PWUP low >=100ms; then high >=50ms before SCCB.
Qualification and second-camera/lighting work remain open.
See ../design/camera_storage_interfaces.md, CMS-014..016.

ORIENTATION SENSOR – WIP
U26: LIS2DW12TR; ST DS11811 Rev 9 pin map.
VDD pin 9 / VDDIO pin 10: +3V3_MCU; CS / SA0 high for I2C address 0x19.
C117/C118: local 100n bypass; GND pins 6/8 and RES pin 7 grounded.
C115: 10u/16V aluminum EEE–FM1C100R; place near VDD pin 9 with C117.
R102/R103: 4.7k SCL/SDA pull–ups; obsolete ADXL VREG network removed.
Host connected: SCL P410/K13; SDA P409/R16; INT1 P306/D11.
Wait 20ms after supply valid. Power–cycle below 100mV for >=10ms.
Pending: sourcing fields, pin modes, timing/power/passive qualification.
See design/sensor_alternative.md; circuit qualification remains open.



Sheet: /Orientation sensor/
File: orientation_sensor.kicad_sch

Title:

Size: A4

Date:

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Rev:

Id: 14/14